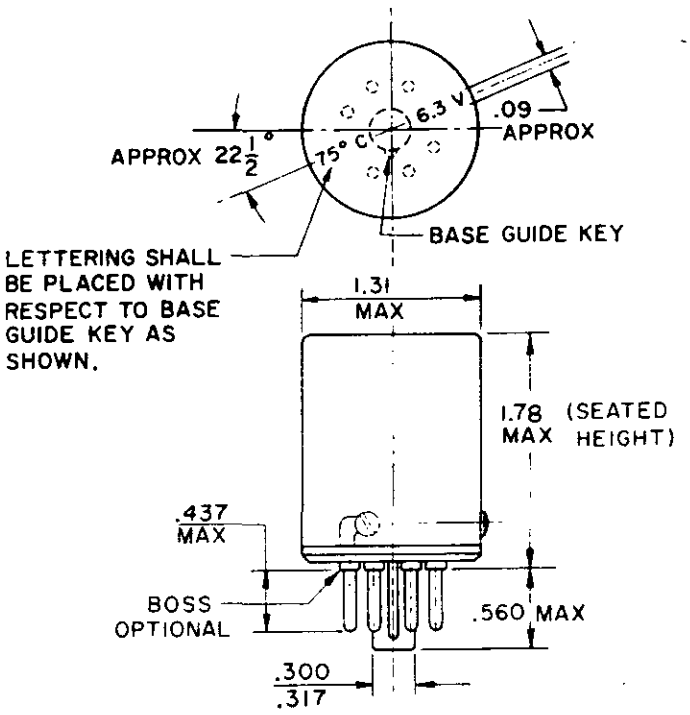
MILITARY SPECIFICATION SHEET

# OVEN, CRYSTAL (+65°C, +75°C, +85°C)

The complete requirements for procuring the ovens described herein shall consist of this document and the latest issue of Specification MIL-O-39021.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

Pertinent characteristics: 2 crystals, HC-6, octal base, 5 watts




| Pins | |
|---|---|
| 2 & 8 | Crystal |
| 4 & 6 | Crystal |
| 5 | Grd. |
| 3 & 7 | Heater |

| INCHES | MM |
|---|---|
| .09 | 2.29 |
| .300 | 7.62 |
| .317 | 8.05 |
| .437 | 11,10 |
| .560 | 14.20 |
| 1.31 | 33.27 |
| 1.78 | 45.21 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. All soldering per MIL-S-6872 using rosin core solder per QQ-S-571.
4. Cover to be secured to base by three fasteners.
5. The oven described under this specification may not be suitable for systems that have EMI requirements in compliance with MIL-STD-461, and shall not be used for new systems.

FSC 5955



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REQUIREMENTS:

Materials:
Case: Anodized aluminum.
Base: Type MFE, MIL-M-14.

Design and construction: See figure.

Physical characteristics:
Weight: 4 ounces maximum.

Oven cavity: Shall accommodate two HC-6/U holders.

Grounding and circuit continuity: See figure.

Terminal configuration and identification: See figure.

Pin alinement: The base shall mate with an octal socket, MIL-S-12883/1.

Rated voltage: See table I.

Dielectric withstanding voltage: For quality conformance inspection, test voltage shall be 400 volts, rms for one second.

Capacitance:

| Between pins | Capacitance, max (pf) |
|---|---|
| 2 and 8 | 1.5 |
| 4 and 6 | 1.5 |

Note: All other pins grounded.

Warmup power: 8.0 watts maximum.

Warmup time:
Conditioning prior to test: 1 hour at -40° ±2°C.
Warmup time: Not to exceed 15 minutes.

Average operating power:
After warmup: Not to exceed 5 watts at 25°C.

Oven cavity temperature:

65° ±2.5°C at room ambient.
Stabilization temperature:
+25°C, -40°C, +25°C, +60°C.
Tolerance for, -40°C and +60°C:
±2°C of oven cavity temperature.

75° ±2.5°C at room ambient.
Stabilization temperature:
+25°C, -40°C, +25°C, +70°C.
Tolerance for, -40°C and +70°C:
±2°C of oven cavity temperature.

85° ±2.5°C at room ambient.
Stabilization temperature:
+25°C, -40°C, +25°C, +80°C.
Tolerance for, -40°C and +80°C:
±2.5°C of oven cavity temperature.

Moisture resistance: Not applicable.

Salt spray (corrosion): Not applicable.

Life: 1 cycle shall consist of 1 hour off and 1 hour on.

Explosion: Ovens shall be tested in accordance with method 109 of MIL-STD-202. The following detail shall apply:
(a) Electrical load - Ovens shall be operated at their maximum rated inductive current, and open circuit voltage.

Part number: Consists of the basic number of this specification sheet and a sequentially assigned dash number.
Example: M39021/2-1.

TABLE I. Dash number and characteristics.

| Rated voltage | Part number M39021/2- | | |
|---|---|---|---|
| | +65°C | +75°C | +85°C |
| 5.7-6.9 | 1 | 5 | 9 |
| 12-14 | 2 | 6 | 10 |
| 24-28 | 3 | 7 | 11 |
| 110-120 | 4 | 8 | 12 |



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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, SM
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - None
Navy - AS, OS, CG
Air Force - 19

Preparing activity:
Army - EL

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